Protocols – PROSS Deseing

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

*This protocol produces a **stabilised variant of an enzyme, designed computationally**, without altering its catalytic function. The PROSS server combines phylogenetic sequence information with Rosetta energy calculations to lower the free energy of the fold, and returns a small set of designs ordered from the most conservative to the most aggressive. The catalytic and cofactor-binding regions are protected before submission, so that no mutation is introduced near the active site.*

*The protocol also covers an **additional step** for transferring a design to a homologous enzyme from a different strain, which cannot be done by simply applying the mutations: the design has to be compared with the new sequence first. The reasons why this transfer was necessary in our project are described in the Experiments page (Computational protein design, section 2).*

Materials list

Input data

- Amino-acid sequence of the target enzyme (FASTA).
- Three-dimensional structure of the enzyme, **in complex with its ligands** (PDB entry, or a model if no structure is available).
- Amino-acid sequence of the homologous enzyme, only for the transfer step.

Software and servers

- **PROSS** server (https://pross.weizmann.ac.il), free academic registration.
- **PyMOL**, to identify the residues surrounding the ligands.
- Sequence alignment tool, only for the transfer step.
- A valid e-mail address, since the server returns the results by e-mail.

Steps

A. Preparing the input:

1. Obtain the amino-acid sequence of the enzyme and check that it matches the sequence of the structure you will use.
2. Download the structure from the PDB. Choose a structure **in complex with its ligands** wherever possible: the complex is what allows the functional residues to be identified, so that they can be excluded from the design.
3. Identify the ligands present in the structure (substrate, substrate analogues, cofactors, metal ions) and note, for each one, its **abbreviation, residue number and chain**, exactly as they appear in the PDB file.

B. Defining the protected regions:

1. Open the structure in PyMOL and display the ligands.
2. Select all residues within **4 Å** of any ligand. These form the catalytic and cofactor-binding shells and must not be mutated.
3. Note the list of selected positions. It is worth keeping an image of them mapped onto the structure, both as a record and for the wiki.

C. Submitting to the PROSS server:

1. Register on the PROSS server and start a new job.
2. Upload the sequence and the structure.
3. Declare the ligands to be protected, giving the abbreviation, position and chain of each one.
4. Submit the job and wait for the results, which arrive by e-mail.

D. Choosing a design:

1. The server returns several designs, numbered from the most conservative to the most aggressive: the higher the number, the more mutations and the greater the predicted stabilisation, but also the greater the risk to catalytic function.
2. Check that none of the designs mutates any of the protected positions identified in step 6.
3. Choose a design according to how much stabilisation is needed and how much risk is acceptable. A conservative design is the safer starting point, since fewer mutations are less likely to disturb the enzyme.
4. Order the selected design as a synthetic gene, together with the **wild-type sequence**, which is the control: because the two differ only in the introduced mutations, any difference between them is attributable to the design.

E. Transferring a design to a different strain background (optional):

A PROSS design is tied to the sequence it was generated from, so it cannot be applied directly to a homologous enzyme from another strain, even one of high sequence identity. If the design has to be moved to a different background, follow these steps:

1. Align the PROSS design, the sequence it was generated from and the natural sequence of the new strain, position by position.
2. Classify every position where the three sequences do not coincide into one of three categories:
   - Positions that **differ naturally** between the two strains.
   - Positions where the **natural residue of the new strain is the one PROSS proposed**. These need no mutation at all, and are the best supported, since the algorithm predicted them and evolution had already fixed them.
   - Positions where PROSS proposes a residue **present in neither natural sequence**. These are the genuinely new mutations, and the ones that carry the real risk.
3. Transfer only the mutations that are both predicted to stabilise and absent from the new sequence.
4. Order the resulting construct as a synthetic gene, with the **unmodified sequence of the new strain** as its control.

Resources & Downloads

- PROSS server: https://pross.weizmann.ac.il
- PyMOL: https://pymol.org
- RCSB Protein Data Bank: https://www.rcsb.org

Troubleshooting

- **The server is unavailable or the job never returns.** The PROSS server was unavailable on our first attempt and the submission had to be repeated, so the design we used comes from a second run. Allow for server downtime when planning this step, and do not leave it to the last minute before a gene order.

- **A design mutates a residue of the active site.** Check that the ligands were declared correctly in the submission, with the right abbreviation, number and chain. If the ligands are not recognised, the server stabilises the whole protein without protecting anything.

- **The most aggressive design loses activity.** More predicted stabilisation is not necessarily better: beyond a certain number of mutations the risk to catalytic function outweighs the gain. Start at the conservative end of the gradient and only move up if the stability obtained is insufficient.

- **No ligand-bound structure is available for your enzyme.** Use an apo structure or a model, but identify the active-site residues from the literature or from a homologous complex, and protect them manually.

- **The design cannot be applied to a related enzyme.** Do not transfer the mutations directly. Follow section E: aligning first avoids introducing mutations that the new enzyme already carries naturally.